

Many companies are coming up with drivers for brushless DC motor-based fans that run directly on battery supply. So this system can be used for such fans as well. In this way, one can save on the cost of an inverter and make lighting and fan systems more efficient and eco-friendly.

The system has two batteries. When both the batteries are charged, load is connected to the first battery (BATT.1) by default. If this battery's voltage goes below the defined level, the load shifts to the second battery and the first battery starts charging. Similarly, when the second battery gets discharged, the load is shifted back to the first battery and the second battery

starts charging. The system also monitors solar voltage. If the voltage is not sufficient for charging, the battery gets charged through the battery charger driven by AC mains.

Circuit and working

Fig. 1 shows the circuit of the solar-powered home lighting system. The circuit is built around microcontroller PIC16F877A (IC1), two 80W solar panels (connected in parallel at CON1), two 12V, 80Ah batteries (BATT.1 and BATT.2) and a 16×2 LCD. A separate 12V battery charger (connected at CON2) is used to charge the batteries when solar voltage is not sufficient. The LED lights made using 12V LED

strips available off-the-shelf can be connected in parallel at CON3.

PIC16F877A is a low-power, high-performance,

CMOS 8-bit microcontroller. Its main features are 8kB Flash, 256 bytes of EEPROM, 368 bytes of RAM, 33 input/output (I/O) pins, 10-bit 8-channel analogue-to-digital converter (ADC), three timers and a watchdog timer with its own on-chip crystal oscillator for reliable operation.

Port pins RB0 through RB7 of IC1 are connected to pins D0 through D7 of LCD1. Port pins RD5 through RD7 of IC1 are connected to RS, R/W and EN of LCD1, respectively. Preset VR1 is used for contrast control of the LCD. The LCD shows information about which battery is charging and the source used. ADC channels RA0, RA1 and RA3 of IC1 monitor the voltage levels of solar panel, BATT.1 and BATT.2, respectively. Presets VR2, VR3 and VR4 are used so that the voltage levels never exceed 5V. Microcontroller IC1 converts these analogue levels into digital form and controls the overall process.

Port pins RC0 and RC1 of IC1 are used to drive relays RL1 and RL2 to select the battery to be charged. If both the batteries are fully charged, BATT.1 gets connected to the load by default. Once this battery gets discharged, the load gets switched to BATT.2 and BATT.1 starts charging.

The source of charging (solar or battery charger) is selected through relay RL3. If the solar voltage is sufficient for charging, port pin RC2 drives relay RL3 to connect the solar charging system to the battery, otherwise it is connected to the battery charger. The battery connected to the load is selected using relays RL4 and RL5 that are driven through port pins RC3 and RC4.

The power supply for the microcontroller is always delivered through BATT.1 after proper regulation. Switch S2 is used to turn on/off the system. A 4MHz crystal is connected to pins 13 and 14 of microcontroller IC1 to provide the basic clock frequency. Power-on reset is provided by the combination of resistor R6 and capacitor C4. Switch S1 is used for manually resetting the microcontroller. Diodes D1 through D5 act as freewheeling diodes.

Configuration Bits set in code

Address	Value	Field	Category	Setting
3007	3F59	OSC	Oscillator	XT
		WDT	Watchdog Timer	Off
		POT	Power Up Timer	Off
		BODEN	Brown Out Detect	Off
		LVF	Low Voltage Program	Disabled
		CPD	Data EE Read Protect	Off
		WRT_PROT	Flash Program Write	Write Protection Off
		CP	Code Protect	Off

Fig. 2: Configuration bits

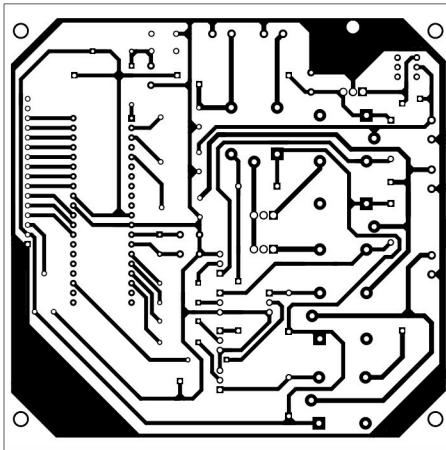


Fig. 3: An actual-size, single-side PCB for the solar-powered home lighting system